



# CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE(Advanced)  
FULL SYLLABUS  
29-04-2022

## JEE(Main + Advanced) : ENTHUSIAST COURSE : (SCORE-I)

### PAPER-2

Time : 3 Hours

Maximum Marks : 180

READ THE INSTRUCTIONS CAREFULLY

#### GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **28** pages and that all the **19** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

#### OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

#### DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take  **$g = 10 \text{ m/s}^2$**  unless otherwise stated.

#### QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

**SOME USEFUL CONSTANTS**

**Atomic No.** : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

**Atomic masses** : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140,

- |                                    |                                                              |
|------------------------------------|--------------------------------------------------------------|
| • Boltzmann constant               | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$                   |
| • Coulomb's law constant           | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$                   |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$  |
| • Speed of light in vacuum         | $c = 3 \times 10^8 \text{ ms}^{-1}$                          |
| • Stefan-Boltzmann constant        | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$                        |
| • Permeability of vacuum           | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$                |
| • Permittivity of vacuum           | $\epsilon_0 = \frac{1}{\mu_0 c^2}$                           |
| • Planck constant                  | $h = 6.63 \times 10^{-34} \text{ J-s}$                       |

Space for Rough Work

**HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS**

**BEWARE OF NEGATIVE MARKING**

**PART-1 : PHYSICS**

**SECTION-I (i) : (Maximum Marks: 24)**

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all ) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

*Full Marks* : +4 If only (all) the correct option(s) is (are) chosen.

*Partial Marks* : +3 If all the four options are correct but **ONLY** three options are chosen.

*Partial Marks* : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

*Partial Marks* : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

*Zero Marks* : 0 If none of the options is chosen (i.e. the question is unanswered).

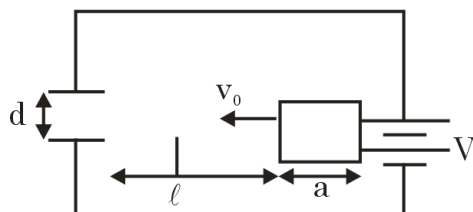
*Negative Marks* : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. In a standard meter bridge experiment to measure the specific resistance of a wire the following data are found.  
 Length (L) =  $(1 \pm 0.01)$  m (measured by meter scale)  
 Radius of cross section of wire =  $(1 \text{ mm} \pm 0.01) \text{ mm}$  (measured by screw gauge)  
 Resistance of wire (R) =  $(5 \pm 0.01) \Omega$   
 Which of the following are possible error in measurement of specific resistance?  
 (A)  $0.65 \mu\Omega\text{m}$                       (B)  $0.44 \mu\Omega\text{m}$                       (C)  $0.28 \mu\Omega\text{m}$                       (D)  $1.25 \mu\Omega\text{m}$
2. A vertical Capillary tube with inner radius of 0.25 mm is submerged into water so that length of its protruding part over the water surface equal to 25 mm. Surface tension of water is  $73 \times 10^{-3} \text{ N/m}$ . Angle of contact is zero degree for glass and water, acceleration due to gravity is  $9.8 \text{ m/s}^2$ . Choose the correct statement. (R = Radius of meniscus and h = height of water in capillary tube.)  
 (A) R = 0.25 mm                      (B) h = 59.6 mm                      (C) R = 0.60 mm                      (D) h = 25 mm

**Space for Rough Work**

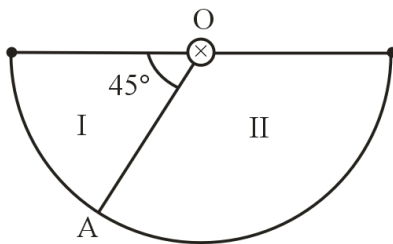
3. A parallel plate capacitor is made of square conducting plates of side 'a' and separation 'd'. The capacitor is connected with battery of emf  $V$  volt as shown. There is dielectric slab of dimension  $a \times a \times d$ , with dielectric constant  $K$ . At  $t = 0$  slab given velocity  $v_0$  towards the capacitor as shown. Neglect gravity and electrostatic force on dielectric when it is outside capacitor and also ignore friction force. Let  $X$  be the length of dielectric inside the capacitor at  $t = t$  ( $\ell \gg a$ ). Here system consists of dielectric and capacitor. (Assume there is no heat loss while dielectric is moving between the plates)



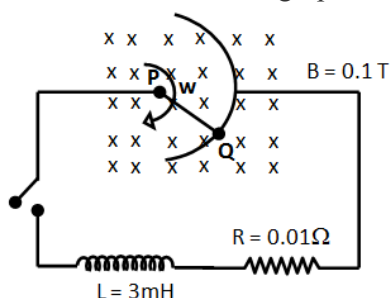
- (A) Motion of dielectric slab is periodic but not SHM  
 (B) Motion of dielectric slab is SHM  
 (C) At any time the slope of graph of total energy of system versus ' $X$ ' is twice the slope of graph of potential energy of system versus ' $X$ '  
 (D) The value of maximum energy stored in the system is  $\frac{1}{2}mv_0^2 + \frac{\epsilon_0 a^2 V^2}{2d}(2K - 1)$
4. A karate master strikes a pine board with a speed of 10 m/s which decreases to 1 m/s in a time interval of 0.002 s. The mass of hand and arm is 1 kg. The thickness of the pine board is 1 cm and volume  $9.0 \text{ cm}^3$ . The breaking stress of the pine board is  $3.6 \times 10^6 \text{ N/m}^2$ . Then,
- (A) force on the board is 3500 N.  
 (B) stress in the board due to the force is  $5.0 \times 10^6 \text{ N/m}^2$ .  
 (C) the board will break, whatever be the mass of the board.  
 (D) the board will not break if thickness is increased.

Space for Rough Work

5. A pendulum swing back forth between two horizontal positions as shown in the figure under the influence of gravity. Line OA divides the whole motion in two parts then:



- (A) There are more than one points in part II where acceleration is vertical  
 (B) There is at least one point in part II where acceleration is horizontal  
 (C) During motion in both parts tension is maximum only once  
 (D) Acceleration of bob never directed towards point of suspension 'O'
6. The circuit shown has resistance  $R = 0.01\Omega$  and inductance  $L = 3\text{ mH}$  connected to a conducting rod PQ of length  $\ell = 2\text{ m}$  which can slide on perfectly conducting circular arc of radius  $\ell$ . with its centre at 'P' Assume friction and gravity to be zero and constant magnetic field  $B = 0.1\text{ T}$  exist as shown. At  $t = 0$  switch is closed and simultaneously external torque is applied about P so that it rotates with uniform angular speed  $\omega = 2\text{ rad/s}$ . Which of the following options are correct about applied torque



- (A) Torque is  $4\text{ N} \cdot \text{m}$  on rod at  $t = 0.3 \ln 2$       (B) Torque is  $8\text{ N} \cdot \text{m}$  on rod at  $t = 0.3 \ln 2$   
 (C) Torque is  $6\text{ N} \cdot \text{m}$  on rod at  $t = 0.6 \ln 2$       (D) Torque is  $12\text{ N} \cdot \text{m}$  on rod at  $t = 0.3 \ln 2$

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

*Full Marks* : +3 If **ONLY** the correct option is chosen.

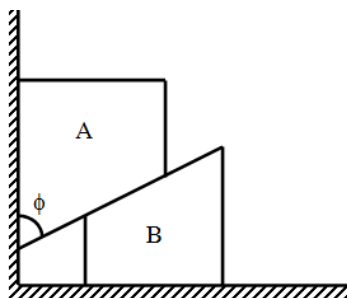
*Zero Marks* : 0 If none of the options is chosen (i.e. the question is unanswered).

*Negative Marks* : -1 In all other cases.

Paragraph for Questions 7 and 8

In the figure given below there are two wedges A and B placed over each other. Wedge A is touching the wall and wedge B is on the floor. Coefficient of friction between A and wall and that between B and floor is  $\mu$ . There is no friction between A and B. Mass of A and B are the same and equal to 9 kg.

[Take  $g = 10 \text{ m/s}^2, \cos\phi = 3/5$ ]



7. Find the minimum value of  $\mu$  required to prevent slipping  
 (A)  $\frac{1}{6}$  (B)  $\frac{1}{4}$  (C)  $\frac{2}{5}$  (D)  $\frac{1}{3}$
8. Find the minimum horizontal force that has to be applied to wedge B in order to move wedge A upwards. (consider  $\mu$  to be the answer obtained in the above problem)  
 (A) 120 N (B) 160 N (C) 180 N (D) 210 N

Space for Rough Work

**Paragraph for Questions 9 and 10**

The carbon in living organism contains about 1.3 atoms of  $C^{14}$  for every  $10^{12}$  atoms of all kinds of carbon. The half life of  $^{14}C$  is 5733 years.

(Take :  $N_A = 6 \times 10^{23} \text{ mol}^{-1}$ , molar mass = 12 g/mol)

9. How many decays would one expect per minute from 2g of carbon taken from a living tree?
- (A)  $\frac{\ln 2}{441} \times 10^{11}$  (B)  $\frac{\ln 2}{2318} \times 10^5$
- (C)  $\frac{\ln 2}{219} \times 10^5$  (D)  $\frac{\ln 2}{3670} \times 10^{12}$
10. If 2g of carbon is extracted from a wooden beam of a prehistoric dwelling and if they undergo 10 decays per minute, about how old is the dwelling?
- (A) 30000 - 35000 years
- (B) 25000 - 30000 years
- (C) 20000 - 25000 years
- (D) 5000 - 10000 years

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**Space for Rough Work**

**SECTION-II (i) : (Maximum Marks: 12)**

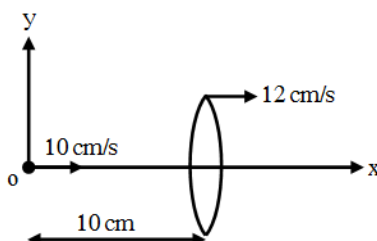
- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

*Full Marks* : +2 If ONLY the correct numerical value is entered at the designated place;

*Zero Marks* : 0 In all other cases.

**Paragraph for Questions 1 and 2**

At  $t = 0$  an object is at origin and lens is at  $x = 10$  cm as shown. Velocity of object and lens is  $10\hat{i}$  cm/s and  $12\hat{i}$  cm/s respectively. initially radius of curvature of equiconvex lens is 10 cm and it is increasing at a rate of 1 cm/s. Assume that refractive index of lens remain same at 1.5.

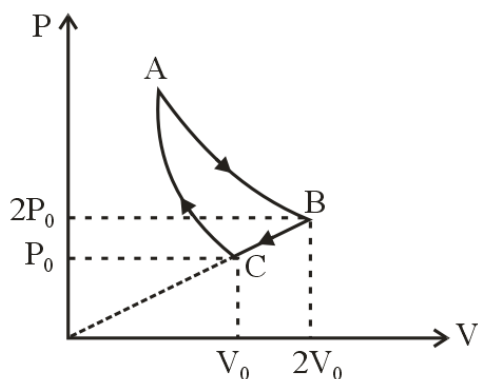


1. X co-ordinate of image is \_\_\_\_\_ cm at  $t = 10$  sec
2. Velocity of image at  $t = 10$  sec is \_\_\_\_\_ m/s

**Space for Rough Work**

### Paragraph for Questions 3 and 4

Mixture of 0.15 mol of He and 0.15 mol of  $O_2$  is used in a cyclic process as shown in P-V graph. Here, process AB is isothermal and CA is adiabatic process.  $P_0$  and volume  $V_0$  are  $10^5$  Pa and 36 liter respectively. [take  $R = \frac{25}{3}$  J/mol  $\times$  K and  $\ln 2 = 0.7$ ]



3. Volume of mixture in state A is \_\_\_\_\_ cc.
4. Total work done in cyclic process is \_\_\_\_\_ kJ.

### Paragraph for Questions 5 and 6

A uniform wire of mass per unit length  $\mu = 1$  g/km is lowered into a tunnel passing through the centre of earth. Length of wire is kept such that separation between centre of mass and centre of gravity of the wire is half of length of wire. Take radius of earth as 6400 km. (Take :  $g = 9.8$  m/s<sup>2</sup>)

5. Length of wire is \_\_\_\_\_ km.
6. Force required to hold wire is \_\_\_\_\_ N.

Space for Rough Work

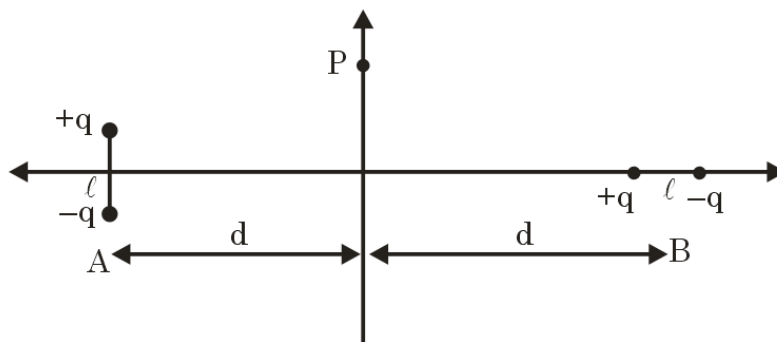
SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

*Full Marks* : +4 If only the correct answer is given.

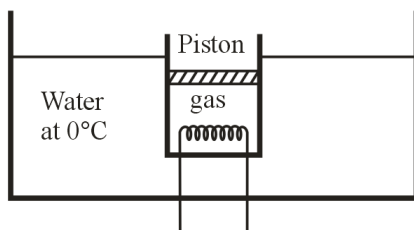
*Zero Marks* : 0 In all other cases

7. Two identical dipoles are placed on the x - axis as shown in the diagram below. Center of each dipole is at a distance  $d$  from the origin. ( $d \gg \ell$ ) if, electric field due to dipole A is parallel to x-axis at a position P on y-axis, then y - component of net electric field at position P is given as  $E_y = \frac{q\ell\sqrt{\beta}}{9\pi\epsilon_0 d^3}$ . Find the value of  $\beta$ .

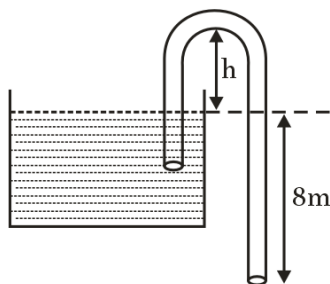


Space for Rough Work

8. An ideal mono atomic gas is filled inside a cylindrical container and it is closed by a smooth and insulating piston of mass 10 kg. The base of cylindrical container is insulating but side walls of cylinder has conductivity of 2 W/mK. Radius of cylinder is  $\frac{10}{\sqrt{\pi}}$  cm, height is 100 cm and thickness of side walls is  $\frac{\sqrt{\pi}}{10}$  cm. The cylindrical container is now placed inside a large water tank as shown. The temperature of water and surrounding is maintained at 0°C. An electrical heater is inserted in the cylinder supplying heat energy at a constant Rate of 4.44 kW. After some time when the temperature of gas reaches 20°C. The rate at which the height of piston rise is  $\alpha$  m/s. Find the value of  $50\alpha$ . Length of gas column at 20°C is 50 cm. (take atmospheric pressure =  $10^5$  Pa)



9. Siphon tube is a narrow pipe that is used to flow water out from a large water tank, as shown. The tube is bent, such that its highest point is at a height  $h$  above the water level and the water out let opening is 8 m below the water level. If the pressure of flowing water at the highest point is never less than the pressure value of 1 m static water height and atmospheric pressure is equal to 10.3 m of static water height. Then find the maximum value of  $10h$  (in m).



Space for Rough Work

**PART-2 : CHEMISTRY**  
**SECTION-I (i) : (Maximum Marks: 24)**

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all ) the correct answer(s)
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*Partial Marks* : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

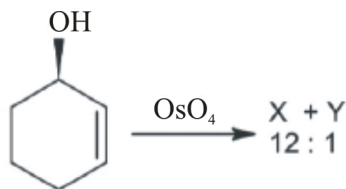
*Partial Marks* : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

*Zero Marks* : 0 If none of the options is chosen (i.e. the question is unanswered).

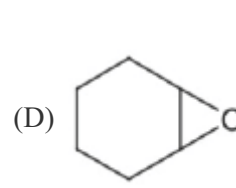
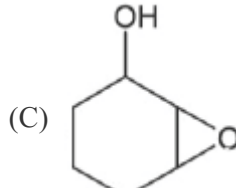
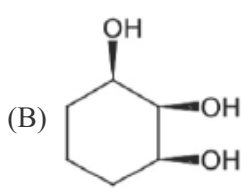
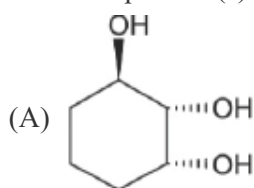
*Negative Marks* : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1.



Mark the product(s) which is/are not X



Space for Rough Work

2. Which of the following statement(s) is/are **CORRECT** ?
- (A) For a **H-atom**, the energies of an electron in 2s and 2p-subshells is the same
- (B) The **maximum number** of electrons possible in a penultimate shell of an atom is 18
- (C) A **d-orbital** can have a maximum of two electrons, with opposite spin.
- (D) In the ground state of a sodium atom all the electrons have the same spin
3. Which of the following statement is/are **INCORRECT**.
- (A) Boron is diagonally related to magnesium.
- (B)  $\text{Bi}^{5+}$  ion has smaller radius compared to  $\text{Bi}^{+3}$
- (C) Last member of the seventh period of periodic table will have atomic number of 124 (if discovered)
- (D) ZnO is an amphoteric oxide
4. Select **CORRECT** statements regarding  $\sigma$  and  $\pi$  bonds.
- (A)  $\sigma$ -bond electron cloud lies on the line joining the nuclei of bonded atoms
- (B)  $\pi$ -electron cloud lies on either side to the line joining the nuclei of bonded atoms
- (C)  $(2p_{\pi} - 3d_{\pi})$   $\pi$ -bond is stronger than  $(2p_{\pi} - 3p_{\pi})$   $\pi$ -bond
- (D)  $\sigma$ -bond has primary effect to decide direction of covalent bond, while  $\pi$ -bond has no primary effect in direction of bond.

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**Space for Rough Work**

5. Which of the following properties are the same for  $N_2H_2$ ,  $N_2F_2$  and  $H_2N_2O_2$  compounds
- (A) All are having two isomers
- (B) All are having equal number of covalent bonds between both N-atoms
- (C) All are having equal bond energy of N-N bond
- (D) Oxidation state of N-atom is same for all compounds
6. Which of the following reaction(s) can occur during the calcination of the involved ore?
- (A)  $Zn(OH)_2ZnCO_3 \xrightarrow{\Delta} 2ZnO + CO_2 \uparrow + H_2O \uparrow$
- (B)  $4FeS_2 + 11O_2 \xrightarrow{\Delta} 2Fe_2O_3 + 8SO_2 \uparrow$
- (C)  $2Al(OH)_3 \xrightarrow{\Delta} Al_2O_3 + 3H_2O \uparrow$
- (D)  $MgCl_2 \cdot 6H_2O \xrightarrow{\Delta} MgCl_2 + 6H_2O \uparrow$

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Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, choose the option corresponding to the correct answer.
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*Full Marks* : +3 If **ONLY** the correct option is chosen.

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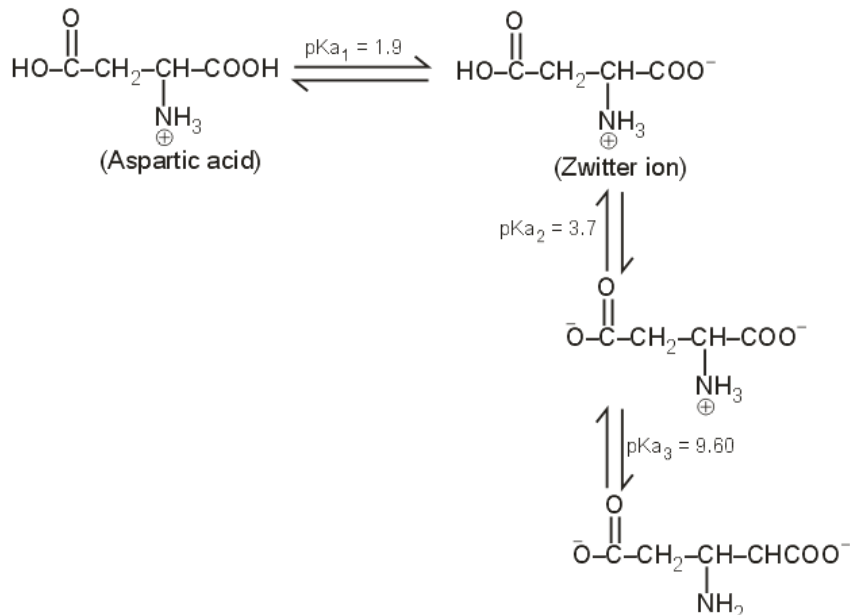
*Negative Marks* : -1 In all other cases.

Paragraph for Questions 7 and 8

The glycine as well as other amino acids, is amphoteric, means it contains an acidic functional group and a basic functional group. The  $-\overset{\oplus}{N}H_3$  is an acidic functional group and basic functional group is –

$COO^-$ . In a strongly acidic medium the species present is  $-\overset{\oplus}{N}H_3CH_2COOH$ . As pH is raised, a proton is removed from this species. Amino acids are crystalline solids that do not melt but on being heated they decompose. These are partially soluble in water but insoluble in non polar organic solvents. These properties are attributed to fact that the stable form of amino acids is a Zwitter ion or inner salt.

7. In following reaction chain



The isoelectric point of aspartic acid is

- (A) 2.8                      (B) 4.8                      (C) 5.75                      (D) 1.9
8. When glycine is dissolved in a solution having pH = 9; at 25°C; it will move towards the electrode 'X' in the electrolytic cell combination. The electrode 'X' is :
- (A) Anode                      (B) Cathode
- (C) Will not move towards any electrode                      (D) None of these

**Paragraph for Questions 9 and 10**

Given

$$E_{\text{Ni}^{2+}/\text{Ni}}^{\circ} = -0.24 \text{ V}$$

$$E_{\text{Sn}^{2+}/\text{Sn}}^{\circ} = -0.12 \text{ V}$$

 A solution containing  $\text{SnSO}_4(1\text{M})$  and  $\text{NiSO}_4(1 \text{ M})$  is electrolyzed. (assume neutral aqueous solution)

9. Which of the ions will be deposited preferentially at cathode? (Given pH of pure water = 7) ( $P_{\text{H}_2} = 1 \text{ atm}$ )
- (A)  $\text{Ni}^{2+}$  (B)  $\text{Sn}^{2+}$   
 (C)  $\text{H}^+$  (D)  $\text{OH}^-$
10. The concentration of  $\text{Sn}^{2+}$  when both  $\text{Ni}^{2+}$  and  $\text{Sn}^{2+}$  starts depositing simultaneously at the cathode.  
 (assume  $\frac{2.303RT}{F} = 0.06$ )
- (A) 1 M (B)  $10^{-4} \text{ M}$   
 (C)  $10^{-3} \text{ M}$  (D)  $10^{-8} \text{ M}$

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**Space for Rough Work**

**SECTION-II (i) : (Maximum Marks: 12)**

- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

*Full Marks* : +2 If ONLY the correct numerical value is entered at the designated place;

*Zero Marks* : 0 In all other cases.

**Paragraph for Questions 1 and 2**

Cadmium oxide crystallizes in NaCl type of crystal lattice. The compound is however usually non-stoichiometric with approximate formula  $\text{CdO}_{0.95}$ . The defect arises due to some cationic positions being occupied by the neutral Cd-atoms instead of  $\text{Cd}^{2+}$  ions and equivalent numbers of anion sites are left vacant : (Given : Cd = 112, O = 16)

1. What percentage of anion sites are vacant ? (in g/cc)
2. If the edge length of the unit cell is 500 pm, what would be the density of perfect solid (in g/cc) ?

**Space for Rough Work**

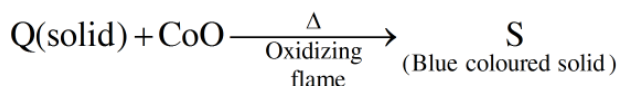
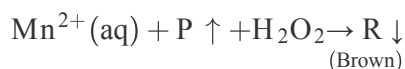
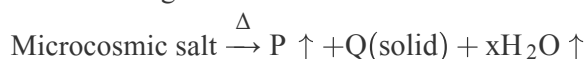
**Paragraph for Questions 3 and 4**

The conductivity of a 0.1 M NaOH solution is  $0.022 \Omega^{-1} \text{cm}^{-1}$ . When an equal volume of 0.1 M HCl solution is added, the conductivity decreases to  $0.0056 \Omega^{-1} \text{cm}^{-1}$ . A further addition of HCl solution, the volume of which is equal to that of first portion added, conductivity increases to  $0.017 \Omega^{-1} \text{cm}^{-1}$ . Calculate molar conductance of NaOH and HCl solutions in the given conditions.

3.  $\Lambda_m$  of NaOH in  $\Omega^{-1} \text{cm}^2 \text{mol}^{-1} =$  \_\_\_\_\_
4.  $\Lambda_m$  of HCl in  $\Omega^{-1} \text{cm}^2 \text{mol}^{-1} =$  \_\_\_\_\_

**Paragraph for Questions 5 and 6**

For following reactions :



Answer following questions :

5. When solid 'R' is fused with KOH in the presence of air, then molecular weight of Mn containing compound formed = \_\_\_\_\_  
(K = 39, Mn = 55, P = 31, Co = 59, Na = 23)
6. Molecular weight of S = \_\_\_\_\_

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**Space for Rough Work**

**SECTION-II (ii) : (Maximum Marks: 12)**

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

*Full Marks* : +4 If only the correct answer is given.

*Zero Marks* : 0 In all other cases

- 
7. How many isomers of  $C_4H_{10}O$  reacts with  $CH_3MgBr$  to evolve  $CH_4$  gas? (Excluding stereoisomer)
  8. Write the sum of geometrical isomers in  $\left[ Pt \left( H_2N - CH(CH_3) - COO^{\ominus} \right)_2 \right]$  complex and stereoisomers of  $[Pt(gly)_3]^{\oplus}$  complex.
  9. 2.47 g of an element  $A_x$  when dissolved in 50 g of benzene, lowered the freezing point of benzene by 0.4 K.  $K_f$  for benzene is  $5.12 \text{ K kg mol}^{-1}$ . The value of x, for atomic weight of A being 79, is [Give your answer as nearest integer]
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**Space for Rough Work**

**PART-3 : MATHEMATICS**  
**SECTION-I (i) : (Maximum Marks: 24)**

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all ) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

*Full Marks* : +4 If only (all) the correct option(s) is (are) chosen.

*Partial Marks* : +3 If all the four options are correct but ONLY three options are chosen.

*Partial Marks* : +2 If three or more options are correct but ONLY two options are chosen and both of which are correct.

*Partial Marks* : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.

*Zero Marks* : 0 If none of the options is chosen (i.e. the question is unanswered).

*Negative Marks* : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. Let  $z_1, z_2, z_3, z_4 \in \mathbb{C}$  (none of  $z_i$  are zero and no two of  $z_i$  are equal for  $i = 1, 2, 3, 4$ ) be such that

$$z_1 (z_1 - z_2) (z_1 - z_3) (z_1 - z_4) = 4$$

$$z_2 (z_2 - z_1) (z_2 - z_3) (z_2 - z_4) = 4$$

$$z_3 (z_3 - z_1) (z_3 - z_2) (z_3 - z_4) = 4$$

$$z_4 (z_4 - z_1) (z_4 - z_2) (z_4 - z_3) = 4$$

Then :

(A)  $z_1 + z_2 + z_3 + z_4 = 0$

(B)  $z_1 z_2 z_3 z_4 = -1$

(C)  $z_1^2 + z_2^2 + z_3^2 + z_4^2 = 0$

(D)  $z_1 z_2 z_3 z_4 = 0$

**Space for Rough Work**

2. Let  $f(x)$  is non negative function defined for  $x \geq 1$  such that  $f'(x) \leq m f(x)$  holds every where in the domain for some positive real number  $m$ . If  $f(1) = 0$ , then

(A)  $f(x)$  is neither even nor odd.

(B)  $\lim_{x \rightarrow 2} (5 + f(x) + x^2 - 4x) \left( \frac{f(2)+e^2}{e^x - e^{2(x-1)+f(x)}} \right)$  is equal to  $e^2$

(C) Number of solution of the equation  $f(x) = e^x - x^2$  is 1

(D)  $\int_{f(e)-1}^{f(e^2)+1} \frac{dx}{2^x + 1}$  is equal to 1

3. Let us consider  $f(x) = \tan x + \frac{\tan 2x}{\operatorname{cosec}^2 x - 1}$  and  $g(n) = \sum_{r=0}^n f(2^r)$

Then which of the following option(s) are correct.

(A) Number of solutions of the equation  $f(x) = 1$  in  $x \in (-\pi, \pi)$  is 2

(B) Number of solutions of the equation  $f(x) = 1$  in  $x \in (-\pi, \pi)$  is 6

(C) If  $g(2020) = \tan(2^m) - \tan(2^r)$ , then  $m + r$  is 2019

(D) If  $g(2020) = \tan(2^m) - \tan(2^r)$ , then  $m + r$  is 2021

4. Let  $P$  is a skew symmetric matrix of order  $3 \times 3$  with real entries. Given  $I + P$  and  $I - P$  are non-singular matrices and  $Q = (I + P)(I - P)^{-1}$  (where  $\det(Q) > 0$ ). Identify the correct statements.

(A)  $\det(2Q) = 8$

(B)  $QQ^T = I$

(C)  $\det(\operatorname{adj} Q) = 1$

(D)  $Q = Q^{-1}$

Space for Rough Work

5. Identify the correct statements

- (A) If  $f(x) > 0$ ,  $f''(x) > 0 \forall x \in [0,1]$  and  $f(0) = 0$ ,  $f(1) = 1$ , then  $f(x) \cdot f^{-1}(x) < x^2 \forall x \in (0,1)$
- (B) If  $f(x)$  is a twice derivable function and  $f(\alpha) = f(\beta) = 0$ ,  $(\alpha < \beta)$  then  $(f'(x))^2 + f(x)f''(x) = 0$  has at least two real roots in  $x \in (\alpha, \beta)$
- (C) Let  $g : [a, b] \rightarrow \mathbb{R}$  be twice differntiable function such that  $g''(x) + g'(x) \cdot g(x) = g(x) \forall x \in [a, b]$  and  $g(a) = g(b) = 0$ , then  $g(x) = 0 \forall x \in [a, b]$
- (D) Let  $f : [-1,1] \rightarrow \mathbb{R}$  be a differentiable function such that  $f(0) = 0$  and  $f(x) > 0 \forall x \in (-1,1) - \{0\}$ , then  $\frac{f'(c)}{f(c)} = \frac{3}{2} \frac{f'(1-3c)}{f(1-3c)} + \frac{1}{1-2c}$  for atleast one  $c \in \left(\frac{1}{3}, \frac{1}{2}\right)$

6. Which of the following is/are correct

- (A) Let  $f(x) = \frac{x}{e^x - 1} + \frac{x}{2} + \log \left( \frac{1 - x + x^2}{1 + x + x^2} \right)^x \forall x \in \mathbb{R} - \{0\}$  is an even function
- (B) Let  $f : \mathbb{R} \rightarrow \mathbb{R}$ ,  $f(x) = \frac{x^3}{3} + \frac{x^2}{2} + ax + b$ , then  $f(x)$  is invertible iff  $a \in \left[\frac{1}{4}, \infty\right)$ ,  $b \in \mathbb{R}$
- (C) The number of real solutions of the equation  $x^3 + 1 = 2\sqrt[3]{2x-1}$  is 3
- (D) The number of solutions of the equation  $z^{61} = \bar{z}$  is K,  $z \in \mathbb{C}$ . Then the last digit of number  $(277)^K$  is 3

Space for Rough Work

**SECTION-I (ii) : (Maximum Marks: 12)**

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

*Full Marks* : +3 If **ONLY** the correct option is chosen.

*Zero Marks* : 0 If none of the options is chosen (i.e. the question is unanswered).

*Negative Marks* : -1 In all other cases.

**Paragraph for Questions 7 and 8**

A variable line  $L$  intersects the parabola  $y = x^2$  at points  $P$  and  $Q$  where  $x$ -coordinates  $\alpha$  and  $\beta$  respectively with  $\alpha < \beta$ , the area of the region enclosed by the segment  $PQ$  and the parabola is always equal to  $\frac{4}{3}$ . The variable segment  $PQ$  has its middle point as  $M$ .

7. Which of the following is correct ?
- (A)  $\beta - \alpha$  can have more than one real values. (B)  $\beta - \alpha$  can be equal to 2
- (C)  $\beta - \alpha$  can have exactly two real value. (D)  $\alpha = 2 + \beta$
8. Equation of pair of tangents are drawn from origin to the locus of  $M$  is
- (A)  $x^2 - y^2 = 0$  (B)  $16x^2 - y^2 = 0$  (C)  $4x^2 - y^2 = 0$  (D) None of these

**Paragraph for Questions 9 and 10**

Let  $f(x) = \int_0^1 e^{|t-x|} dt$ ,  $0 \leq x \leq 1$  then

9. Maximum value of  $f(x)$  equals to
- (A)  $e - 1$  (B)  $2(\sqrt{e} - 1)$  (C)  $2\sqrt{e} - 1$  (D)  $\sqrt{e} - 2$
10. Minimum value of  $f(x)$  equal to
- (A)  $e - 1$  (B)  $\sqrt{e} - 1$  (C)  $2(\sqrt{e} - 1)$  (D)  $2\sqrt{e} - 1$

**Space for Rough Work**

**SECTION-II (i) : (Maximum Marks: 12)**

- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

*Full Marks* : +2 If ONLY the correct numerical value is entered at the designated place;

*Zero Marks* : 0 In all other cases.

**Paragraph for Questions 1 and 2**

$f(x)$  is defined for  $x > -1$  and has a continuous and twice derivable function. If  $f$  satisfies

$f(0) = 1$ ,  $f'(0) = 0$  and  $(1 + f(x)) f''(x) = 1 + x$

1. The maximum value of  $f'(2)$  is equal to
2. The maximum value of  $f(2)$  is equal to  $K$ , then  $[K]$  is ( $\because [.]$  greatest integer function)

**Space for Rough Work**

**Paragraph for Questions 3 and 4**

If  $f: \mathbb{R} \rightarrow \mathbb{R}$ ,  $f(x) = 4x - \cos \pi x$ ;  $g: \mathbb{R} \rightarrow \mathbb{R}$ ,  $g(x) = x^3 + 4x - \cos \pi x$ . Then

3.  $\frac{d}{dx}(g^{-1} \circ f^{-1}(x))$  at  $x = -3$  is  $\frac{1}{k^2}$  then  $|k|$  is
4.  $\frac{d}{dx}(f^{-1} \circ g^{-1}(x))$  at  $x = -4$  is  $\frac{1}{\lambda}$  then  $\frac{\lambda}{4}$  is equal to

**Paragraph for Questions 5 and 6**

A fair coin is tossed  $n$  times probability for ' $r$ ' heads  $= \frac{{}^nC_r}{2^n}$  on the basis of information.

5. Probability of at least five consecutive heads occur from a coin tossed 10 times is  $p/q$ , then  $p + q$  is (where  $p$  &  $q$  are relatively coprime)
6. Probability of at least four consecutive tails occur from a coin tossed 9 times is  $\frac{a}{b}$ , then  $a + b$  is (where  $a$  &  $b$  are relatively coprime)

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**Space for Rough Work**

**SECTION-II (ii) : (Maximum Marks: 12)**

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

*Full Marks* : +4 If only the correct answer is given.

*Zero Marks* : 0 In all other cases

7. Let  $\vec{a}, \vec{b}, \vec{c}, \vec{d}$  are four distinct unit vectors in space, such that  $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{c} = \vec{a} \cdot \vec{c} = \vec{b} \cdot \vec{d} = \vec{c} \cdot \vec{d} = -\frac{1}{11}$ .  
The value of  $\vec{a} \cdot \vec{d}$  can be expressed as  $-\frac{p}{q}$ , where p & q are co-prime positive integer. Then the value of p + q is
8. If the value of  $\sum_{r=2}^{\infty} \cot^{-1}(r^2 - 5r + 7)$  is equal to  $\frac{a\pi}{b}$  (where a and b are coprime), then the value of (a + b) is
9. Let there are 6 hats of different colours and 6 shirts of same colours as that of hats. If the number of ways in which these can be put on by 5 men such that no man wear the shirt and the hat of the same colour is  $K \times 6!$ . Then the value of K is.

**Space for Rough Work**

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Space for Rough Work

NAME OF THE CANDIDATE .....

FORM NO. ....

I have read all the instructions  
and shall abide by them.

\_\_\_\_\_  
Signature of the Candidate

I have verified the identity, name and Form  
number of the candidate, and that question  
paper and ORS codes are the same.

\_\_\_\_\_  
Signature of the Invigilator

Space for Rough Work